

Books and Reading Guide

A Structured Study Roadmap for Image Denoising with Residual Networks

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1 How to Use This Guide

1.1 What this document is, and what it is not

This is a **study roadmap**, not a bibliography. The companion document, *Deep Residual Learning for Image Denoising* (referred to throughout as **the technical document**), derives the mathematics behind the two notebooks in the `Image_denoising_using_ResNet` repository and cites 54 research papers. Papers are the right medium for a specific result, and a poor medium for learning a subject: they assume the background rather than supply it, they compress derivations, and they are written for people who already know why the problem matters.

Books supply what papers omit. This guide identifies **14 books** that between them cover every mathematical object the technical document uses, and for each one it states:

- the exact bibliographic identity (edition, publisher, year, ISBN, DOI or publisher page), verified against the publisher’s own record;
- **why it is relevant to this project specifically** — not to image processing in general;
- **which chapters to read**, and which to skip;
- **what to extract** from those chapters — the specific result, derivation, or intuition that pays off;
- **how it connects to the actual code** in the two notebooks;
- its **level** and its **classification** as foundational, core, supplementary, or advanced;
- its **relationship to the papers** already cited in the technical document — which citation it provides the background for.

A note on chapter numbering: it can shift between editions and printings. Where a chapter number is given below it corresponds to the edition named in that entry; readers using a different edition should confirm against their own copy’s table of contents, which is why each chapter reference is given with its title as well as its number.

1.2 The gap this project sits in

The repository trains two convolutional residual networks to remove additive white Gaussian noise from grayscale LFW face images. Understanding that work properly requires four distinct bodies of knowledge, and no single book covers more than two of them:

1. **Where noise comes from and what it is.** Sensor physics, the Poisson–Gaussian model, and the idealization to AWGN. (Technical document Sections 1.2 and 2.1.)
2. **What “estimating a clean image” means.** MAP versus MMSE estimation, ill-posedness, regularization, and Tweedie’s formula — which is what makes the global residual skip connection the *principled* architecture rather than a trick. (Sections 2.2–2.4, 3.3.)
3. **What the classical methods were, and why they stopped improving.** Spatial filtering, anisotropic diffusion, total variation, wavelet shrinkage, non-local means, BM3D. (Section 4.)
4. **How convolutional networks work and why residual learning and channel attention help.** (Sections 3.3–3.4, 5, 6, 7.)

Reading only deep-learning material teaches you to build the enhanced notebook without understanding why its architecture takes the form it does. Reading only classical material leaves you unable to read the code. The reading order in Section 2 is designed to close both gaps in the right sequence.

1.3 Three routes through the material

Not everyone needs all fourteen books. Pick a route.

Route A — “I need to understand and extend this code” (about 6 weeks). Kok & Tam (Ch. 1–3, 5–6) → Goodfellow et al. (Ch. 5–9) → Prince (Ch. 10–11) → Wang & Bovik. Four books, minimum theory, maximum direct relevance. This route gets you to the point where every line of both notebooks is intelligible and you can run the ablation grid proposed in Section 12.1 of the technical document.

Route B — “I want the theory properly” (one semester). Route A, preceded by Kay (Ch. 10–12) and followed by Chan & Shen (Ch. 4), Mallat (Ch. 11), and Hansen (Ch. 1–5). Seven books. This is the route that makes the technical document’s Sections 2 and 4 feel obvious rather than asserted.

Route C — “I am starting research in image restoration” (two semesters). All fourteen, in the order given in Section 2, with Aubert & Kornprobst, Elad, and Barbu read as reference works rather than cover to cover.

2 Reading Order at a Glance

#	Book (short)	Tier	Read for	Technical-doc sections
1	Gonzalez & Woods, <i>Digital Image Processing</i>	Foundational	Vocabulary, noise models, spatial filtering, PSNR	1.2, 3.1, 3.5, 4.1
2	Kay, <i>Estimation Theory</i>	Foundational	MAP, MMSE, the Bayesian frame	2.2, 2.3, 7.1
3	Kok & Tam, <i>Digital Image Denoising in MATLAB</i>	Core	The whole classical pipeline, implemented	4 (all)
4	Goodfellow et al., <i>Deep Learning</i>	Core	Convolution, optimization, regularization, BN	3.1, 3.4, 7.5–7.6
5	Prince, <i>Understanding Deep Learning</i>	Core	Residual networks, attention, diffusion	3.3, 5.4–5.6, 6

#	Book (short)	Tier	Read for	Technical-doc sections
6	Wang & Bovik, <i>Modern Image Quality Assessment</i>	Core	SSIM, from the people who invented it	3.6, 7.3–7.4
7	Chan & Shen, <i>Image Processing and Analysis</i>	Core	TV/ROF, PDE and wavelet denoising unified	4.2–4.4
8	Mallat, <i>A Wavelet Tour</i>	Core	Wavelet shrinkage, minimax optimality	4.4
9	Hansen, <i>Discrete Inverse Problems</i>	Supplementary	Ill-posedness, regularization, parameter choice	2.4
10	Boyd & Vandenberghe, <i>Convex Optimization</i>	Supplementary	Proximal operators, projection, convexity	3.5, 4.4, 7
11	Szeliski, <i>Computer Vision</i>	Supplementary	Where denoising sits in a vision pipeline	1.2, 4, 5
12	Elad, <i>Sparse and Redundant Representations</i>	Advanced	Sparsity, K-SVD, the BM3D worldview	4.5–4.6
13	Aubert & Kornprobst, <i>Mathematical Problems in Image Processing</i>	Advanced	Rigorous PDE/variational analysis	4.2–4.3
14	Barbu, <i>Novel Diffusion-Based Models</i>	Advanced	Modern nonlinear diffusion variants	4.2

The tiers are cumulative: nothing in Tier “Core” assumes anything not in Tier “Foundational”, and the Advanced entries assume the Core.

3 Tier 1 — Foundational

These two books establish the vocabulary and the estimation-theoretic frame. Read them first; everything afterwards assumes them.

3.1 Gonzalez & Woods, *Digital Image Processing*

Field	Value
Authors	Rafael C. Gonzalez, Richard E. Woods
Edition	4th edition
Publisher	Pearson
Year	2018
ISBN	978-0-13-335672-4
Pages	1192
Level	Undergraduate / early graduate
Classification	Foundational

Why it is relevant to this project. This is the book that defines the terms the technical document uses without ceremony: image as a sampled and quantized 2-D function, intensity normalization to $[0, 1]$, spatial filtering as correlation with a kernel, padding, and the taxonomy of noise. Before you can appreciate that the notebooks' `add_gaussian_noise` implements one specific and rather idealized corruption model, you need to have seen the others — Rayleigh, gamma, exponential, uniform, and impulse (salt-and-pepper) — laid out with their densities side by side. Gonzalez & Woods is where that comparison lives, and it is also where the median filter is developed, which matters because the notebooks apply a Gaussian-trained network to salt-and-pepper noise and report 22.56 dB without comparing against the matched estimator (technical document Section 10.3).

Topics it covers that appear in the technical document.

- Noise probability density functions and how to estimate noise parameters from a flat image patch (technical document Sections 1.2 and 8.2, where the network must infer σ internally).
- Spatial filtering: linear smoothing, the bias–variance trade-off of averaging, and order-statistic filters (Section 4.1).
- Frequency-domain view of smoothing and why noise occupies the high band (Section 3.3.2, the argument for residual learning).
- Restoration in the presence of noise only, and Wiener filtering (the classical ancestor of BM3D's second stage, Section 4.6).
- PSNR and MSE as fidelity measures (Section 3.5).

Chapters to study.

- **Chapter 2, *Digital Image Fundamentals*** — sampling, quantization, intensity, and the coordinate conventions. Skim if already comfortable.
- **Chapter 3, *Intensity Transformations and Spatial Filtering*** — read carefully. This is the mathematical content behind `Conv2D`, including correlation versus convolution (the technical document's convolution equation is a correlation, as Gonzalez & Woods explains), kernel separability, and padding.
- **Chapter 4, *Filtering in the Frequency Domain*** — read the sections on lowpass and highpass filtering. This gives the spectral intuition for why an image's energy is concentrated at low frequencies while AWGN is flat, which is the core of the residual-learning argument.
- **Chapter 5, *Image Restoration and Reconstruction* — the most important chapter in this book for this project.** Read the noise-model section, the spatial-filtering-for-noise-only section (mean filters, order-statistic filters, adaptive filters), and the Wiener filtering section.

- **Chapter 7, *Wavelet and Other Image Transforms*** — read as a gentle preparation for Mallat.
- Skip: compression, morphology, segmentation, and representation chapters, none of which touch this project.

What to extract. Three things. (i) The full table of noise models, so that “AWGN” registers as *one choice among many*. (ii) The derivation of why an $m \times m$ mean filter divides noise standard deviation by m while blurring edges — this is the tension every method in the technical document’s Section 4 tries to resolve. (iii) The adaptive local-noise-reduction filter, which modulates its strength by the ratio of local variance to noise variance; it is the classical antecedent of what a blind CNN denoiser must learn to do implicitly.

Connection to the code. `images_clean = color.rgb2gray(...)` and the normalization to $[0, 1]$ are Chapter 2 material. The 3×3 convolutions with `padding='same'` are Chapter 3. `add_salt_and_pepper_noise` and the evaluation in cell 18/19 are Chapter 5.

Relationship to the cited papers. Provides the background for essentially every classical citation in the technical document’s Section 4, and specifically prepares the reader for Perona & Malik (1990) and Rudin, Osher & Fatemi (1992) by first establishing what a linear filter can and cannot do.

3.2 Kay, *Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory*

Field	Value
Author	Steven M. Kay
Volume	Volume I: Estimation Theory
Publisher	Prentice Hall
Year	1993
ISBN	978-0-13-345711-7
Pages	595
Level	Graduate
Classification	Foundational

Why it is relevant to this project. The technical document’s Section 2 claims that denoising *is* an estimation problem, that MAP and MMSE are two different answers to it, and that the MMSE estimator is the posterior mean — which is then used in Section 7.1 to explain why MSE-trained denoisers blur. Those are not image-processing facts; they are estimation-theory facts, and Kay is the standard reference that proves them. If you read one thing to make the technical document rigorous rather than plausible, read Kay’s Bayesian chapters.

Topics it covers that appear in the technical document.

- The classical/Bayesian distinction, and why a prior is *required* when the likelihood alone has no useful maximum (technical document Section 2.4).
- MMSE estimation and the theorem that the minimum mean-square-error estimator is the conditional mean $\mathbb{E}[x | y]$ (the MMSE equation). This is the single most load-bearing result borrowed from outside image processing in the whole technical document.

- MAP estimation and its relationship to MMSE (they coincide for Gaussian posteriors, and diverge exactly where the posterior is skewed or multimodal — which is the blur argument of Section 7.1).
- Linear Bayesian estimation and the Wiener filter, which is what BM3D’s second stage computes (Section 4.6).
- The bias–variance decomposition and the notion of an estimator’s risk.

Chapters to study.

- **Chapter 10, *The Bayesian Philosophy*** — read in full. It sets up prior, posterior, and the difference between treating x as deterministic-unknown versus random.
- **Chapter 11, *General Bayesian Estimators* — the essential chapter.** It derives the MMSE estimator as the posterior mean, the MAP estimator as the posterior mode, and compares them. Everything in the technical document’s Sections 2.2, 2.3 and 7.1 is a specialization of this chapter to $y = x + n$.
- **Chapter 12, *Linear Bayesian Estimators*** — the LMMSE/Wiener material. Read at least the scalar and vector Gaussian cases.
- **Chapters 2–4** (minimum variance unbiased estimation, the Cramér–Rao lower bound, linear models) — read if you want the classical half of the picture, particularly the CRLB as a statement about fundamental limits, which is the estimation-theoretic cousin of Levin & Nadler’s (2011) denoising bounds.
- **Chapter 7, *Maximum Likelihood Estimation*** — useful context; explains why $\hat{x} = y$ is the ML answer for the denoising likelihood and hence why ML alone is useless here.
- Skip: the signal-processing application chapters (spectral estimation, array processing) unless they interest you independently.

What to extract. The proof that $\arg \min_f \mathbb{E} \|x - f(y)\|^2 = \mathbb{E}[x | y]$, and the *intuition* for what a posterior mean does when the posterior is multimodal. Once you have internalized that a squared-error-optimal estimator must average over all plausible reconstructions, the blurriness of MSE-trained networks stops being a mysterious empirical fact and becomes a theorem.

Connection to the code. The baseline notebook’s `loss='mean_squared_error'` is a decision to compute $\mathbb{E}[x | y]$. The enhanced notebook’s `combined_loss` is a decision to compute something else on purpose. Kay is where you learn what that difference means.

Relationship to the cited papers. Supplies the background for Robbins (1956), Miyasawa (1961) and Efron (2011) — the Tweedie’s-formula lineage in Section 2.3 — and for the Blau & Michaeli (2018) perception–distortion trade-off, which is a statement about the geometry of estimators that only makes sense once Kay’s Chapter 11 is understood.

4 Tier 2 — Core

These five books are the heart of the guide. Together they cover the classical denoising pipeline, the deep-learning machinery, and the quality metrics.

4.1 Kok & Tam, *Digital Image Denoising in MATLAB*

Field	Value
Authors	Chi-Wah Kok, Wing-Shan Tam

Field	Value
Publisher	Wiley–IEEE Press
Year	2024
ISBN	978-1-119-61769-3
DOI	10.1002/9781119617778
Level	Advanced undergraduate / graduate, implementation-oriented
Classification	Core — the single most on-topic book in this guide

Why it is relevant to this project. This is the only book in the list whose subject is *exactly* the subject of the repository. Its seven chapters march through the same sequence as the technical document’s Section 4 — filtering, wavelets, variational methods, non-local means — and it supplies runnable MATLAB source for every algorithm, which means you can generate the classical baselines that the notebooks conspicuously lack. The technical document’s Section 11.8 lists “no classical or literature baselines” as a limitation; this book is the most direct route to fixing it.

Verified table of contents.

Ch.	Title	Pages
1	Digital Image	1–46
2	Filtering	47–77
3	Wavelet	79–115
4	Rank Minimization	117–136
5	Variational Method	137–147
6	NonLocal Means	149–168
7	Random Sampling	169–182

Chapters to study, and what each maps onto.

- **Chapter 1, *Digital Image*** — image representation, noise models, and quality metrics. Read alongside the technical document’s Sections 2.1 and 3.5.
- **Chapter 2, *Filtering*** — linear and non-linear spatial filtering. Maps onto technical document Section 4.1. Extract: the quantitative noise-reduction-versus-blur trade-off, and the median filter, which is the correct comparison for the salt-and-pepper experiment in the notebooks’ cell 18/19.
- **Chapter 3, *Wavelet*** — transform-domain thresholding. Maps onto Section 4.4. Extract: a working soft-thresholding implementation, and the practical business of threshold selection, which the technical document treats only theoretically via the universal threshold $\sigma\sqrt{2\log N}$.
- **Chapter 4, *Rank Minimization*** — low-rank models (WNNM and relatives). This is the one classical family the technical document does **not** cover, and it is worth reading precisely for that reason: grouping similar patches into a matrix and shrinking its singular values is a close cousin of BM3D’s collaborative filtering, and it was competitive with early CNN denoisers.
- **Chapter 5, *Variational Method*** — the ROF/TV material. Maps onto Section 4.3. Extract: a numerical solver for the ROF problem, since the technical document gives the Euler–Lagrange equation but not an algorithm.

- **Chapter 6, *NonLocal Means*** — maps onto Section 4.5. Extract: the practical parameterization (search-window size, patch size, filtering parameter h) and the $2\sigma^2$ distance correction discussed in the technical document.
- **Chapter 7, *Random Sampling*** — acceleration strategies; read if the $O(HWS^2P^2)$ cost of NLM is a practical concern for you.

What to extract overall. A set of working classical baselines and a feel for their parameter sensitivity. The most valuable exercise this book enables: run NLM, TV, and wavelet shrinkage on the same 1,000 LFW test images at $\sigma = 0.09$ and put their PSNR/SSIM into the technical document’s results table alongside the 21.42/28.32/33.71 dB rows. Only then do the network numbers acquire absolute meaning.

Connection to the code. Chapter 1’s metric definitions correspond directly to the notebooks’ PSNR and SSIM helper functions; Chapter 6 corresponds to the self-similarity intuition that a CNN’s receptive field only partially replicates.

Relationship to the cited papers. It is the implementation companion to Buades et al. (2005), Rudin, Osher & Fatemi (1992), Donoho & Johnstone (1994), and Dabov et al. (2007) — four of the technical document’s core classical citations.

4.2 Goodfellow, Bengio & Courville, *Deep Learning*

Field	Value
Authors	Ian Goodfellow, Yoshua Bengio, Aaron Courville
Publisher	MIT Press
Year	2016
ISBN	978-0-262-03561-3
Free online	https://www.deeplearningbook.org
Level	Graduate
Classification	Core

Why it is relevant to this project. Every non-architectural decision in the enhanced notebook — Adam, He initialization, batch normalization, the regularizing effect of data augmentation, early stopping, the reason mini-batch size interacts with gradient noise — is treated here with more care than any paper gives it. The technical document states these things; Goodfellow et al. justify them.

Topics it covers that appear in the technical document.

- Convolution as an operation: sparse interactions, parameter sharing, and equivariance to translation. This is precisely the argument the technical document’s Section 3.1 uses to license patch training.
- Optimization for deep models: why SGD-family methods work, learning-rate schedules, and adaptive methods including Adam (Section 7.6).
- Regularization: weight decay, early stopping *as* regularization, dataset augmentation, and noise robustness — the last of these is directly the enhanced notebook’s resampled-noise strategy (Section 8.2).
- Batch normalization, presented as a reparameterization (Section 3.4).

- Maximum likelihood, MSE, and cross-entropy as instances of the same principle, which frames the loss-function discussion of Section 7.

Chapters to study.

- **Chapter 5, *Machine Learning Basics*** — read the sections on capacity, overfitting, the bias–variance trade-off, maximum likelihood, and Bayesian statistics. This is the bridge from Kay to neural networks.
- **Chapter 6, *Deep Feedforward Networks*** — architecture design, hidden units (including why ReLU), and back-propagation.
- **Chapter 7, *Regularization for Deep Learning*** — read in full. Sections on dataset augmentation, noise robustness, and early stopping are all directly instantiated in the notebooks.
- **Chapter 8, *Optimization for Training Deep Models*** — read in full. Covers parameter initialization strategies (including the variance-preserving argument behind `he_normal`), adaptive learning rates and Adam, and batch normalization.
- **Chapter 9, *Convolutional Networks*** — read in full. Convolution, pooling, variants, and the structured-output setting that denoising belongs to.
- Skip on a first pass: Part III (deep generative models) except as noted under Prince below; the practical-methodology chapter is worth a skim.

What to extract. The parameter-initialization argument (why variance $2/\text{fan_in}$ for ReLU), the treatment of *noise injection as regularization*, and a clear understanding of why mini-batch gradient estimates have variance that depends on batch composition — which is the technical document’s Section 8.1 argument that a batch of random crops is a better gradient estimator than a batch of near-identical portraits.

Connection to the code. `kernel_initializer='he_normal'`, `keras.optimizers.Adam`, `BatchNormalization()`, `EarlyStopping`, and the augmentation inside `augment_and_noise` are all chapters 7–9 material.

Relationship to the cited papers. Provides the textbook treatment behind Ioffe & Szegedy (2015), Kingma & Ba (2015), and He et al. (2015). It predates and therefore does not cover ResNets in depth — for that, use Prince.

4.3 Prince, *Understanding Deep Learning*

Field	Value
Author	Simon J. D. Prince
Publisher	MIT Press
Year	2023
ISBN	978-0-262-04864-4
Pages	544
Free draft	https://udlbook.github.io/udlbook/
Level	Graduate, but unusually readable
Classification	Core

Why it is relevant to this project. This is the modern complement to Goodfellow et al., and it covers the three things that book cannot: **residual networks**, **attention**, and **diffusion models**.

Those are, respectively, the architecture of both notebooks, the mechanism of the Squeeze-and-Excitation block, and the reason the technical document’s Tweedie’s-formula discussion matters beyond denoising. It is the single best textbook treatment of the material in the technical document’s Sections 3.3, 5.4–5.6 and 6.

Topics it covers that appear in the technical document.

- Residual connections: why they ease optimization, the shattered-gradients view, and their interaction with batch normalization. Directly Section 3.3.
- Batch normalization revisited with the modern (loss-landscape) explanation rather than the internal-covariate-shift story — matching the technical document’s Section 3.4 caveat.
- Attention and transformers, which is the general form of which Squeeze-and-Excitation is the cheapest special case (Section 6.3).
- Diffusion models, whose training objective is a denoising regression and whose connection to denoising runs through exactly the identity in Section 2.3.

Chapters to study.

- **Chapter 10, *Convolutional Networks*** — read for the modern presentation and for receptive-field reasoning, which the technical document uses quantitatively in Section 3.2.
- **Chapter 11, *Residual Networks* — the essential chapter for this project.** It develops residual connections, explains the gradient-flow argument that the technical document gives as the gradient-flow equation, and covers batch normalization in the residual context.
- **Chapter 12, *Transformers*** — read the attention sections. You are looking for the general pattern: compute a data-dependent weighting, apply it multiplicatively. Once that is clear, SE reads as “attention with a globally pooled query and a diagonal output”, and Restormer’s channel attention reads as the full-rank version.
- **Chapter 18, *Diffusion Models*** — read after the technical document’s Section 2.3. This is where “a denoiser is a score estimator” becomes a constructive algorithm rather than an observation.
- Chapters 1–9 serve as a self-contained replacement for Goodfellow et al. Chapters 5–8 if you prefer a single, more recent source.

What to extract. The intuition for *why* the identity path matters, in a form you can reproduce at a whiteboard: unrolled, a residual network is a sum over paths, and the gradient decomposes into a direct term plus a correction, so no product of Jacobians can drive it to zero.

Connection to the code. `Add()([shortcut, x])` inside `res_block` and `rcab`, and `Add(dtype='float32')([inp, x])` at the model’s output, are Chapter 11 material. `Multiply()([x, s])` inside `se_block` is Chapter 12 material in miniature.

Relationship to the cited papers. The textbook treatment behind He et al. (2016), He et al. (2016b), Vaswani et al. (2017), and Ho et al. (2020).

4.4 Wang & Bovik, *Modern Image Quality Assessment*

Field	Value
Authors	Zhou Wang, Alan C. Bovik
Series	Synthesis Lectures on Image, Video, and Multimedia Processing

Field	Value
Publisher	Morgan & Claypool
Year	2006
ISBN	978-1-59829-022-6
Pages	~108
Level	Graduate; short and self-contained
Classification	Core

Why it is relevant to this project. The enhanced notebook’s loss function is $0.8(1 - \text{SSIM}) + 0.2\ell_1$. SSIM is therefore not a metric this project merely reports — it is the objective the network is trained to optimize, which makes understanding its construction, its invariances, and its failure modes a prerequisite rather than a nicety. This ~100-page monograph is written by SSIM’s inventors and covers the material more completely than the original 2004 paper.

Topics it covers that appear in the technical document.

- The critique of MSE and PSNR as perceptual measures — the argument summarized in the technical document’s Section 3.5.
- The structural-similarity philosophy: separating luminance, contrast, and structure, and the axioms each comparison function must satisfy (Section 3.6.1).
- The full derivation of the SSIM index, including why $C_3 = C_2/2$ collapses the three-factor product into the familiar closed form.
- Multi-scale SSIM (MS-SSIM), which is what Zhao et al. (2017) actually used in the loss mixture the notebook approximates with single-scale SSIM.
- The windowing question — Gaussian versus uniform — which is exactly the implementation discrepancy the technical document flags in Section 3.6.3, where the training loss uses TensorFlow’s 11×11 Gaussian window and the reported metric uses scikit-image’s 7×7 uniform default.

Chapters to study. The book is short enough to read end to end in a sitting or two, and that is the recommendation. If time is very limited, read the chapter developing the SSIM index and the chapter on multi-scale extension, and skip the information-theoretic metrics (VIF and relatives) on a first pass.

What to extract. Two specifics. (i) That the structure term is a Pearson correlation coefficient, hence invariant to affine intensity changes — which is why an SSIM-only loss can let absolute brightness drift, and therefore why the ℓ_1 anchor term in the notebook’s `combined_loss` is not optional. (ii) That SSIM is a *local* index averaged over windows, so the choice of window changes the number; this is what makes the technical document’s Section 3.6.3 warning material rather than pedantic.

Connection to the code. `tf.image.ssim(y_true, y_pred, max_val=1.0)` in `combined_loss` and `skimage.metrics.structural_similarity` in the SSIM helper are two different instantiations of this book’s index. Reading it is what lets you notice that they are not the same thing.

Relationship to the cited papers. The extended treatment of Wang et al. (2004) and Wang et al. (2003), and essential background for Zhao et al. (2017), whose loss function this project adapts.

4.5 Chan & Shen, *Image Processing and Analysis*

Field	Value
Authors	Tony F. Chan, Jianhong (Jackie) Shen
Subtitle	Variational, PDE, Wavelet, and Stochastic Methods
Publisher	Society for Industrial and Applied Mathematics (SIAM)
Year	2005
ISBN	978-0-89871-589-7
Pages	xxii + 400
Level	Graduate, mathematically serious
Classification	Core

Why it is relevant to this project. The technical document’s Section 4 presents four classical families — diffusion, total variation, wavelet shrinkage, and non-local patch methods — as a sequence of increasingly sophisticated priors. Chan & Shen is the book that presents them as *one subject* rather than four, unified by the variational formulation $\min_u \frac{1}{2}\|u - y\|^2 + \lambda R(u)$ that the technical document uses as its organizing equation. Its denoising chapter treats linear filtering, TV, and wavelet methods under a single roof, which is exactly the perspective that makes the technical document’s regularizer table (Section 2.2) meaningful rather than a list.

It is also the direct source for the ROF model’s proper mathematical setting: functions of bounded variation, why the ℓ_1 gradient penalty admits jump discontinuities while a quadratic penalty does not, and what “staircasing” is.

Topics it covers that appear in the technical document.

- Variational image models and the Bayesian/MAP interpretation of regularization (Sections 2.2 and 4.3).
- Total variation denoising: the ROF model, its Euler–Lagrange equation, the role of curvature motion, and the staircasing artifact (Section 4.3).
- PDE methods, including the heat equation as Gaussian smoothing and nonlinear diffusion (Sections 4.1–4.2).
- Wavelet-domain methods and their relationship to variational ones — specifically, that soft thresholding is the proximal operator of an ℓ_1 penalty, which is the connection the technical document draws in Section 4.4.
- Stochastic and MRF image models, which is the probabilistic language in which “learned prior” eventually makes sense.

Chapters to study.

- **Chapter 2, *Some Modern Image Analysis Tools*** — the mathematical toolkit: BV functions, wavelets, and the calculus of variations as used here. Read the BV material carefully; it is what makes the TV-preserves-edges claim precise.
- **Chapter 3, *Image Modeling and Representation*** — how priors are constructed. Read for the conceptual frame.
- **Chapter 4, *Image Denoising* — the essential chapter.** Read in full and in parallel with the technical document’s Section 4.
- Chapters 5–7 (deblurring, inpainting, segmentation) — read only if those problems interest you; deblurring is the natural next inverse problem after denoising and is a good demonstration of why a denoiser is a reusable prior (technical document Section 12.6).

What to extract. The unified variational viewpoint, and specifically the understanding that choosing a denoising algorithm *is* choosing a prior. Once that lands, the deep-learning move — amortize the prior into network weights instead of writing it down — reads as a natural continuation of the same programme rather than as a break with it.

Connection to the code. Indirect but deep: this book explains what the network’s weights have implicitly learned. It is also the theoretical companion to Kok & Tam’s Chapter 5.

Relationship to the cited papers. The textbook treatment of Rudin, Osher & Fatemi (1992) and Perona & Malik (1990), and background for Donoho & Johnstone (1994).

4.6 Mallat, *A Wavelet Tour of Signal Processing: The Sparse Way*

Field	Value
Author	Stéphane Mallat
Edition	3rd edition
Publisher	Academic Press (Elsevier)
Year	2008
ISBN	978-0-12-374370-1
Pages	832
Level	Graduate to research
Classification	Core (for the denoising chapter)

Why it is relevant to this project. The technical document’s Section 4.4 makes three claims about wavelet shrinkage that it does not prove: that an orthonormal transform preserves white Gaussian noise; that the universal threshold $\sigma\sqrt{2\log N}$ is the right order of magnitude because the maximum of N standard normals concentrates near $\sqrt{2\log N}$; and that the resulting estimator is within a logarithmic factor of an oracle. Mallat proves all three. This is also the canonical source for the sparsity principle that underlies BM3D’s collaborative filtering.

Topics it covers that appear in the technical document.

- Orthonormal bases and the invariance of white noise under orthogonal transforms (Section 4.4).
- Nonlinear approximation and the relationship between sparsity and approximation error — the reason a piecewise-smooth image has few large wavelet coefficients.
- Thresholding estimators: hard versus soft, risk analysis, and the oracle comparison.
- Minimax optimality over smoothness classes, i.e. the precise sense in which Donoho & Johnstone’s result is a *guarantee*.
- Translation-invariant denoising / cycle spinning, which explains the pseudo-Gibbs ringing the technical document mentions.

Chapters to study.

- **Chapter 9, *Approximations in Bases*** — read the nonlinear-approximation sections. This is the mathematical content of “images are sparse in a wavelet basis”.
- **Chapter 11, *Denoising* — the essential chapter.** Diagonal estimation in a basis, thresholding risk, the universal threshold, minimax optimality, and translation-invariant variants.

- Chapters 6–7 (wavelet zoom, wavelet bases) — read as needed to support Chapters 9 and 11; the singularity/Lipschitz-regularity material in Chapter 6 explains *why* edges produce large coefficients across scales.
- Chapters 1–5 — reference material; do not read linearly.
- Skip on a first pass: compression, audio processing, and the geometric representation chapters.

What to extract. The oracle argument. Wavelet shrinkage works because there exists an ideal estimator that knows which coefficients carry signal, and thresholding approximates it to within a $\log N$ factor without that knowledge. This is the classical era’s cleanest statement of what a good prior buys you, and it is the right yardstick against which to think about what a CNN’s learned prior buys instead.

Connection to the code. No direct correspondence — the notebooks contain no wavelets. The connection is conceptual and important: a convolutional network’s first layer is a learned filter bank, and the technical document’s argument that the network learns to separate signal from noise by *magnitude in a learned representation* is the same argument, with the basis learned rather than designed.

Relationship to the cited papers. The definitive treatment of Donoho & Johnstone (1994), Donoho (1995), and Mallat (1989); essential background for Dabov et al. (2007), whose 3-D collaborative filtering is a sparsity argument in a data-adaptive grouping.

5 Tier 3 — Supplementary

Useful, but each addresses a specific gap rather than the main line.

5.1 Hansen, *Discrete Inverse Problems: Insight and Algorithms*

Field	Value
Author	Per Christian Hansen
Series	Fundamentals of Algorithms
Publisher	SIAM
Year	2010
ISBN	978-0-89871-696-2
Level	Graduate, computational
Classification	Supplementary

Why it is relevant to this project. The technical document asserts in Section 2.4 that denoising is ill-posed and that regularization is what makes it determinate. Hansen is the book that makes “ill-posed” quantitative: singular value decay, the discrete Picard condition, noise amplification along small-singular-value directions, and the trade-off curve between data fidelity and solution regularity. Denoising is the easiest possible instance (the forward operator is the identity), so Hansen’s machinery is more than the project needs — which is exactly why it is clarifying, because it shows what the *general* problem looks like and hence what denoising is a special case of.

Topics it covers that appear in the technical document.

- Ill-posedness in Hadamard’s sense and its discrete manifestation (Section 2.4).

- Tikhonov regularization, which is the $R(x) = \lambda \|\nabla x\|_2^2$ row of the technical document’s regularizer table.
- Choosing the regularization parameter — the L-curve and generalized cross-validation. This is the classical analogue of the question the enhanced notebook answers by training blind: how strong should the prior be, given that the right answer depends on the noise level?

Chapters to study. Chapters 1–3 for the setup and the SVD analysis of ill-posedness; Chapter 4 for regularization methods; Chapter 5 for parameter-choice methods. The iterative-regularization material later in the book is optional here.

What to extract. The regularization-parameter-selection problem, and the recognition that a *blind* denoiser must solve an amortized version of it: it has to infer the appropriate strength from the input, which is precisely the argument in the technical document’s Section 8.2 for why blind training forces the network to estimate σ internally.

Relationship to the cited papers. Background for the general inverse-problem framing that Venkatakrishnan et al. (2013) exploit in Plug-and-Play priors, and context for Levin & Nadler (2011).

5.2 Boyd & Vandenberghe, *Convex Optimization*

Field	Value
Authors	Stephen Boyd, Lieven Vandenberghe
Publisher	Cambridge University Press
Year	2004
ISBN	978-0-521-83378-3
Free online	https://stanford.edu/~boyd/cvxbook/
Level	Graduate
Classification	Supplementary

Why it is relevant to this project. Three specific results in the technical document are convex-analysis facts. First, that soft thresholding is the proximal operator of the ℓ_1 norm (Section 4.4). Second, that clipping to $[0, 1]$ is Euclidean projection onto a convex set, and that such projections are non-expansive — which is the technical document’s rigorous explanation for why the measured noisy-input PSNR (21.42 dB) exceeds the theoretical 20.92 dB, and why `np.clip` on predictions can only help (Section 3.5). Third, that the MAP objective is convex for the classical regularizers and hence has a unique minimizer, which is what makes those methods well-defined.

Chapters to study. Chapter 2 (convex sets — including projection onto a set), Chapter 3 (convex functions), Chapter 4 (convex optimization problems), and Section 8.1 for the projection material specifically. Chapter 5 (duality) is worth reading if you intend to work with TV solvers, where the dual formulation is how the problem is actually solved. The interior-point chapters are not needed for this project.

What to extract. Proximal operators and projections as *objects*: once prox and P_C are familiar, a large fraction of classical image restoration collapses into a small number of recognizable moves, and the Plug-and-Play idea (replace a proximal operator with a denoiser) becomes an obvious thing to try rather than a clever trick.

Connection to the code. `np.clip(..., 0.0, 1.0)` in `add_gaussian_noise` and in every evaluation cell is $P_{[0,1]^N}$.

5.3 Szeliski, *Computer Vision: Algorithms and Applications*

Field	Value
Author	Richard Szeliski
Edition	2nd edition
Series	Texts in Computer Science
Publisher	Springer
Year	2022
ISBN	978-3-030-34371-2
DOI	10.1007/978-3-030-34372-9
Pages	947
Free draft	https://szeliski.org/Book/
Level	Advanced undergraduate to graduate
Classification	Supplementary

Why it is relevant to this project. Szeliski is the book that puts denoising in context: it is one operator inside an imaging pipeline, and the choices made upstream (sensor, demosaicing, colour transform, compression) determine what the noise actually looks like by the time a denoiser sees it. That is precisely the issue behind the technical document’s Sections 9.2 and 11.6 — LFW’s “clean” targets are JPEG-compressed photographs that already contain real sensor noise and demosaicing artifacts, so the experiment measures inversion of synthetic AWGN applied on top of an imperfect reference. The 2nd edition also has a deep-learning chapter, so it bridges the classical and modern halves of the technical document in a single volume.

Chapters to study. The image-formation chapter for the sensor pipeline and photon noise; the image-processing chapter for linear and non-linear filtering (including bilateral filtering, the missing link between Gaussian smoothing and non-local means in the technical document’s Section 4); the model-fitting and optimization chapter for regularization, MRFs and variational methods; and the deep-learning chapter for a compact modern survey. The computational-photography material covers practical denoising and multi-image fusion.

What to extract. The end-to-end pipeline picture, and the bilateral filter, which the technical document mentions only in passing but which is conceptually the halfway house between local averaging and NLM: it weights neighbours by both spatial *and* intensity proximity, making it the first widely used content-adaptive averaging scheme.

Relationship to the cited papers. Background for Tomasi & Manduchi (1998), and general context for the real-noise citations, Plotz & Roth (2017) and Guo et al. (2019).

6 Tier 4 — Advanced and Specialist

Read these when the Core tier is comfortable, or consult them as references.

6.1 Elad, *Sparse and Redundant Representations*

Field	Value
Author	Michael Elad
Subtitle	From Theory to Applications in Signal and Image Processing
Publisher	Springer
Year	2010
ISBN	978-1-4419-7010-7
DOI	10.1007/978-1-4419-7011-4
Level	Graduate to research
Classification	Advanced

Why it is relevant to this project. This is the definitive treatment of the sparse-representation view of denoising: model a patch as $x \approx D\alpha$ with α sparse, and denoise by finding the sparsest representation consistent with the data. It is the row of the technical document’s regularizer table (Section 2.2) labelled “Sparse coding / K-SVD”, and it is the intellectual context in which BM3D was invented. Crucially for this project, it is also the clearest statement of the *learned dictionary* idea — the first move from hand-designed to data-adapted priors, and therefore the direct conceptual predecessor of learning the prior with a CNN.

Topics it covers that appear in the technical document. Sparsity as a prior; ℓ_0 versus ℓ_1 and why the convex relaxation is justified; pursuit algorithms (matching pursuit, basis pursuit); dictionary learning via K-SVD; and patch-based image denoising built on all of the above.

Chapters to study. The theoretical first part (uniqueness of sparse solutions, pursuit algorithms and their guarantees) can be read selectively — the results matter more than the proofs for this project’s purposes. The applications part is what to read carefully: the chapters on dictionary learning and on image denoising, where the K-SVD denoising algorithm is developed. Read the denoising application alongside the technical document’s Section 4.6.

What to extract. The patch-modelling paradigm, and the recognition that *every* strong classical denoiser — NLM, BM3D, K-SVD — is a patch method. The technical document’s Section 4.7 argues that the classical plateau came from the difficulty of extending patch size; Elad’s book is where you see why patch size is the binding constraint.

Relationship to the cited papers. The book-length treatment of Elad & Aharon (2006), and the natural companion to Dabov et al. (2007).

6.2 Aubert & Kornprobst, *Mathematical Problems in Image Processing*

Field	Value
Authors	Gilles Aubert, Pierre Kornprobst
Subtitle	Partial Differential Equations and the Calculus of Variations
Edition	2nd edition
Series	Applied Mathematical Sciences, vol. 147
Publisher	Springer

Field	Value
Year	2006
ISBN	978-0-387-32200-1
DOI	10.1007/978-0-387-44588-5
Level	Research; assumes real analysis and functional analysis
Classification	Advanced

Why it is relevant to this project. The technical document states in Section 4.2 that the Perona–Malik equation is *ill-posed* because its diffusion coefficient across strong edges, $g(s) + s g'(s)$, becomes negative — running the heat equation backwards. It states in Section 4.3 that the ROF model is well-posed in the space of functions of bounded variation. Both are genuine theorems, and this is the book that proves them. Read it if you want the variational and PDE sections of the technical document on a rigorous footing rather than an intuitive one.

Topics it covers that appear in the technical document. Existence and uniqueness for the ROF minimization problem in BV; the mathematical pathologies of the Perona–Malik model and the regularized (Catté–Lions–Morel) variants that repair them; viscosity solutions; and the general calculus-of-variations machinery (direct method, lower semicontinuity, relaxation) that image models require.

Chapters to study. The mathematical-preliminaries chapter as a reference for BV, Sobolev spaces, and the direct method; then the image-restoration chapter, which is where the denoising and deblurring models are analysed. The edge detection, segmentation and sequence-analysis chapters are outside this project’s scope.

What to extract. Precision about a claim the technical document makes informally: that TV “preserves edges” is not a heuristic but a consequence of the fact that BV functions admit jump discontinuities while H^1 functions do not.

Relationship to the cited papers. The rigorous treatment behind Perona & Malik (1990) and Rudin, Osher & Fatemi (1992).

6.3 Barbu, *Novel Diffusion-Based Models for Image Restoration and Interpolation*

Field	Value
Author	Tudor Barbu
Series	Signals and Communication Technology
Publisher	Springer
Year	2019 (hardcover 2018)
ISBN	978-3-319-93005-3 (hardcover); 978-3-030-06566-9 (softcover)
DOI	10.1007/978-3-319-93006-0
Level	Research monograph
Classification	Advanced / specialist

Why it is relevant to this project. A focused monograph on nonlinear diffusion for restoration —

the family the technical document introduces in Section 4.2 through Perona & Malik. Where Aubert & Kornprobst give the classical analysis, Barbu surveys and develops *modern variants*: second- and fourth-order anisotropic diffusion models, hybrid schemes, and their numerical discretizations, with restoration and interpolation as the driving applications.

Why it is worth reading despite the deep-learning turn. Two reasons specific to this project. First, the technical document notes that Chen & Pock’s (2017) trainable nonlinear reaction diffusion unrolls exactly these PDE models into a learnable feed-forward network — so understanding the diffusion models is understanding what that architecture is a learned version of. Second, “diffusion” in the modern generative sense (Ho et al., 2020) is a *different* object that shares a name; reading a book on classical image diffusion is a good inoculation against conflating the two, a confusion that is common and unhelpful.

Chapters to study. Read selectively: the chapters developing the nonlinear diffusion denoising models and their numerical schemes are the relevant ones; the interpolation and inpainting material is adjacent rather than central here. Treat it as a survey to sample rather than a textbook to work through.

What to extract. A sense of how far the hand-designed-PDE programme was pushed, and its performance ceiling — which is the empirical backdrop to the technical document’s Section 4.7 argument about why the field moved to learned priors.

Relationship to the cited papers. Extends Perona & Malik (1990); provides the model class that Chen & Pock (2017) learn.

7 Coverage Matrix: Technical Document to Books

The following table maps each substantive section of the technical document to the book chapters that cover it. Use it to read the two documents together.

Technical-doc section	Topic	Primary source	Secondary
1.2	Sensor noise,	Gonzalez & Woods	Szeliski (image
	Poisson–Gaussian	Ch. 5	formation)
2.1	AWGN observation	Kok & Tam Ch. 1	Gonzalez & Woods
	model		Ch. 5
2.2	MAP estimation,	Kay Ch. 10–11	Chan & Shen Ch. 3–4
	regularizer table		
2.3	MMSE, Tweedie’s	Kay Ch. 11	Prince Ch. 18
	formula		
2.4	Ill-posedness	Hansen Ch. 1–3	Aubert & Kornprobst
			(prelims)
3.1	Convolution,	Goodfellow Ch. 9	Gonzalez & Woods
	equivariance		Ch. 3
3.2	Receptive field	Prince Ch. 10	Goodfellow Ch. 9
3.3	Residual learning,	Prince Ch. 11	Goodfellow Ch. 8
	gradient flow		
3.4	Batch normalization	Prince Ch. 11	Goodfellow Ch. 8
3.5	PSNR; clipping as	Wang & Bovik	Boyd & Vandenberghe
	projection		Ch. 2, §8.1

Technical-doc section	Topic	Primary source	Secondary
3.6	SSIM derivation	Wang & Bovik	Kok & Tam Ch. 1
4.1	Spatial filtering	Gonzalez & Woods Ch. 3, 5	Kok & Tam Ch. 2
4.2	Anisotropic diffusion	Chan & Shen Ch. 4	Aubert & Kornprobst; Barbu
4.3	Total variation / ROF	Chan & Shen Ch. 4	Kok & Tam Ch. 5; Aubert & Kornprobst
4.4	Wavelet shrinkage	Mallat Ch. 11	Kok & Tam Ch. 3
4.5	Non-local means	Kok & Tam Ch. 6	Szeliski (filtering)
4.6–4.7	BM3D; why classical plateaued	Elad (applications)	Kok & Tam Ch. 4
5	Research evolution	Prince Ch. 10–12	Szeliski (deep learning)
6	RCAB, Squeeze-and- Excitation	Prince Ch. 11–12	Goodfellow Ch. 9
7.1–7.4	MSE blur, ℓ_1 , SSIM loss	Kay Ch. 11 + Wang & Bovik	Goodfellow Ch. 5
7.5–7.6	Cosine schedule, Adam, mixed precision	Goodfellow Ch. 8	Prince Ch. 6–7
8	Patches, blind σ , augmentation	Goodfellow Ch. 7	Prince Ch. 9
9	Code walkthrough	(no book; read the notebooks)	Kok & Tam for classical analogues
10	Experiments and results	Wang & Bovik (metrics)	Kok & Tam Ch. 1
11	Limitations	Hansen Ch. 5 (parameter choice)	Szeliski (real pipelines)
12	Future directions	Prince Ch. 18 (diffusion)	Elad (sparsity), Szeliski

7.1 Where the books run out

Three parts of the technical document have **no adequate book coverage**, and must be learned from the papers themselves:

1. **Channel attention specifically** (Sections 6.3–6.4). Prince’s transformer chapter gives the general attention pattern, but Squeeze-and-Excitation and RCAB are covered only in Hu et al. (2018) and Zhang et al. (2018). Both papers are short and readable; read them directly.
2. **The modern restoration state of the art** (Section 5.6). NAFNet (Chen et al., 2022) and Restormer (Zamir et al., 2022) postdate every book here except Prince and Szeliski, neither of which covers them. Read the papers.
3. **Self-supervised denoising** (Section 12.3). Noise2Noise, Noise2Void, and Neighbor2Neighbor have no textbook treatment at all. The Noise2Noise argument is a two-line consequence of Kay’s Chapter 11, however: if $\mathbb{E}[y' \mid x] = x$, then regressing onto y' has the same minimizer as regressing onto x .

8 Suggested Study Plans

8.1 Plan A — Six weeks, code-focused

Week	Reading	Deliverable
1	Gonzalez & Woods Ch. 2–3, 5	Reimplement <code>add_gaussian_noise</code> and a mean/median filter from scratch; verify the 20.92 dB prediction
2	Kok & Tam Ch. 1–3	Run wavelet shrinkage on the LFW test set; record PSNR/SSIM
3	Goodfellow Ch. 7–9	Annotate every layer of <code>build_model</code> with its parameter count; confirm 373,057
4	Prince Ch. 10–11	Derive the receptive fields (25 and 37) independently; verify by ablating blocks
5	Wang & Bovik (whole)	Reproduce both SSIM variants; quantify the Gaussian-vs-uniform window gap
6	Kok & Tam Ch. 5–6	Add TV and NLM rows to the results table; write up

8.2 Plan B — One semester, theory-focused

Weeks 1–2: Kay Ch. 10–12. Weeks 3–4: Gonzalez & Woods Ch. 3–5 and Kok & Tam Ch. 1–3. Weeks 5–6: Chan & Shen Ch. 2–4. Weeks 7–8: Mallat Ch. 9, 11. Weeks 9–10: Goodfellow Ch. 7–9. Weeks 11–12: Prince Ch. 10–12. Weeks 13–14: Wang & Bovik, plus Hansen Ch. 4–5. Final weeks: run the ablation grid from the technical document’s Section 12.1 and write it up.

8.3 Plan C — Research preparation

Plan B, then: Elad (applications), Aubert & Kornprobst (restoration chapter), Barbu (selectively), Prince Ch. 18, and the paper list in the technical document’s Sections 5.6 and 12. At that point the natural project is the one the technical document identifies as most valuable — a properly controlled ablation separating the six confounded changes between the baseline and enhanced notebooks, on a fixed seed and a leakage-free split.

9 Summary Table

Book	Year	ISBN	Tier	Essential chapters
Gonzalez & Woods, <i>Digital Image Processing</i> , 4th ed.	2018	978-0-13-335672-4	Foundational	2, 3, 4, 5, 7
Kay, <i>Estimation Theory</i> , Vol. I	1993	978-0-13-345711-7	Foundational	10, 11, 12
Kok & Tam, <i>Digital Image Denoising in MATLAB</i>	2024	978-1-119-61769-3	Core	1–7 (all)
Goodfellow et al., <i>Deep Learning</i>	2016	978-0-262-03561-3	Core	5, 6, 7, 8, 9
Prince, <i>Understanding Deep Learning</i>	2023	978-0-262-04864-4	Core	10, 11, 12, 18
Wang & Bovik, <i>Modern Image Quality Assessment</i>	2006	978-1-59829-022-6	Core	all (~108 pp.)
Chan & Shen, <i>Image Processing and Analysis</i>	2005	978-0-89871-589-7	Core	2, 3, 4
Mallat, <i>A Wavelet Tour</i> , 3rd ed.	2008	978-0-12-374370-1	Core	9, 11
Hansen, <i>Discrete Inverse Problems</i>	2010	978-0-89871-696-2	Supplementary	1–5
Boyd & Vandenberghe, <i>Convex Optimization</i>	2004	978-0-521-83378-3	Supplementary	2, 3, 4, §8.1
Szeliski, <i>Computer Vision</i> , 2nd ed.	2022	978-3-030-34371-2	Supplementary	image formation, processing, optimization, deep learning applications part
Elad, <i>Sparse and Redundant Representations</i>	2010	978-1-4419-7010-7	Advanced	
Aubert & Kornprobst, <i>Mathematical Problems in Image Processing</i> , 2nd ed.	2006	978-0-387-32200-1	Advanced	preliminaries, restoration

Book	Year	ISBN	Tier	Essential chapters
Barbu, <i>Novel Diffusion-Based Models</i>	2019	978-3-319-93005- 3	Advanced	diffusion denoising models

Fourteen books. Four of them (Kok & Tam, Goodfellow et al., Prince, Wang & Bovik) are sufficient to understand and extend the repository. The remaining ten are what turn that understanding into the ability to judge whether the approach was the right one.